

International Islamic University Islamabad

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Complex Engineering Problem (CEP)

Computer Vision Project

**Project Name: Vision-Based Autonomous
Perception System**

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List of Abbreviations

YOLO	You Only Look Once
FPS	Frames Per Second
CNN	Convolutional Neural Network
TFLite	TensorFlow Lite
NMS	Non-Maximum Suppression

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Chapter 1

System Modeling and Problem Analysis

1.1 Introduction

Autonomous driving systems rely heavily on computer vision to perceive and understand the surrounding environment in real time. Vision-based perception enables autonomous vehicles to detect lane boundaries, recognize obstacles, and make safe navigation decisions. This project focuses on the design and implementation of an autonomous perception system that performs lane detection and object detection using deep learning and computer vision techniques, with real-time visualization and embedded deployment considerations.

1.2 Problem Analysis

Designing a real-time perception system for autonomous driving presents several engineering challenges:

- Variations in lighting conditions such as daytime, nighttime, and shadows
- Image noise and motion blur caused by vehicle movement
- Limited computational and memory resources on embedded platforms
- Strict real-time processing constraints for safe decision-making

These challenges require an efficient, optimized, and robust vision pipeline capable of operating under real-world conditions.

1.3 System Overview

The proposed autonomous perception system consists of the following major components:

- Camera module for capturing real-time road scene images or video streams
- Preprocessing and deep learning models for lane and object detection

- Control and decision logic for managing thresholds and performance
- Real-time visualization interface using Streamlit UI
- Embedded deployment on Raspberry Pi for edge inference

System Architecture and Workflow

The overall system operation can be understood through two complementary perspectives. The first diagram presents a **high-level system workflow**, while the second diagram illustrates a **detailed end-to-end processing pipeline** including internal control logic.

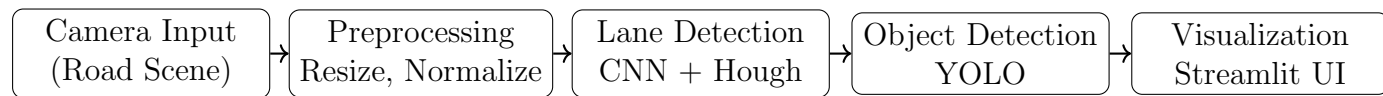


Figure 1.1: High-Level System Workflow from Image Acquisition to Visualization

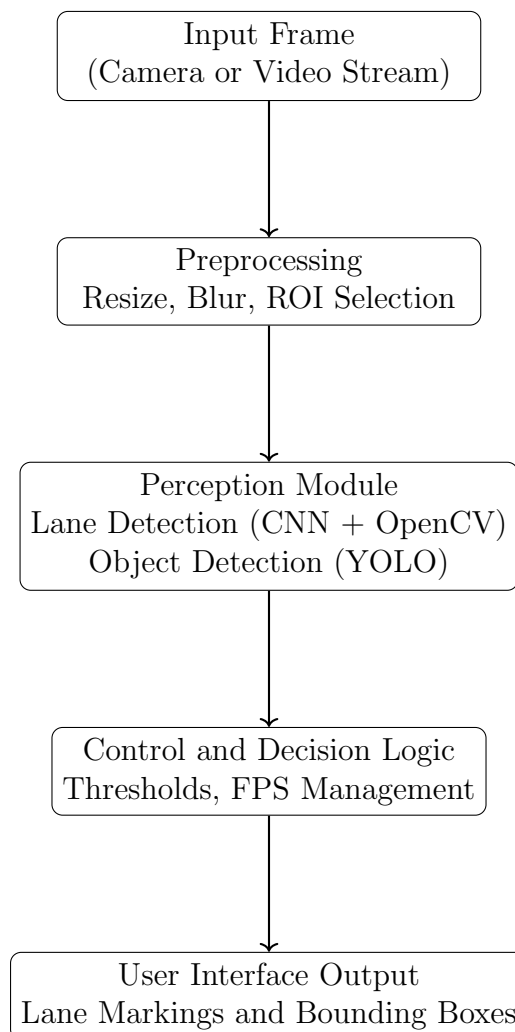


Figure 1.2: End-to-End Processing Workflow of the Autonomous Perception System

Chapter 2

Dataset Preparation and Preprocessing

2.1 Dataset Description

Public road scene datasets and recorded traffic videos were used for training and testing. Frames were extracted from videos to form the dataset.

2.2 Preprocessing Steps

- Image resizing and normalization
- Grayscale conversion for lane detection
- Noise reduction using Gaussian blur
- Dataset splitting into training and validation sets



Figure 2.1: Sample frames from traffic dataset used for training and evaluation

Chapter 3

Deep Learning Model Design and Training

3.1 Lane Detection Pipeline

Lane detection uses image preprocessing and Hough Transform to identify lane lines:

- Convert image to grayscale
- Apply Gaussian blur to remove noise
- Detect edges using Canny detector
- Select region of interest
- Apply Hough Transform to detect lane lines

3.2 CNN for Lane Detection

A CNN model was trained to enhance lane detection accuracy:

- Input: $128 \times 128 \times 3$ image
- Layers: Convolution + ReLU, Max Pooling
- Flatten and fully connected layer to predict lane mask

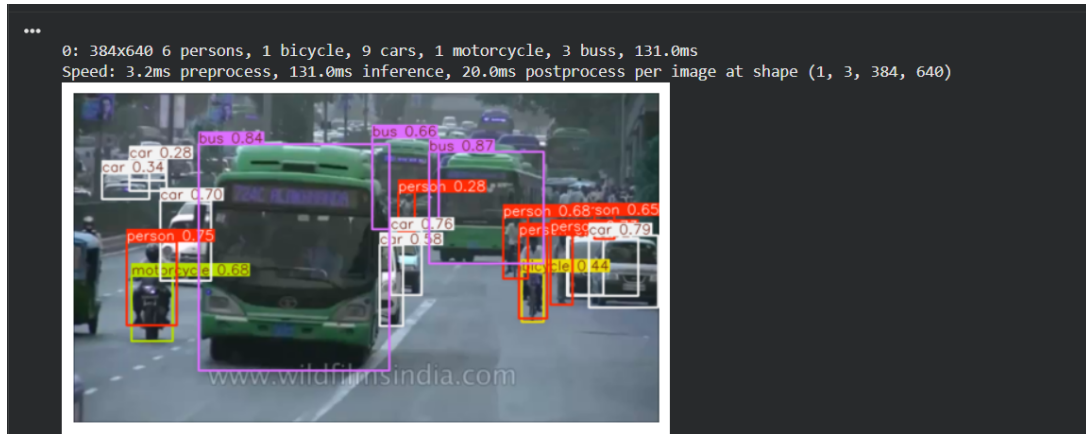


Figure 3.1: Lane detection output on sample road scene

Chapter 4

Object Detection Using YOLO

4.1 Introduction to YOLO

You Only Look Once (YOLO) is a real-time object detection system that predicts bounding boxes and class probabilities in a **single evaluation**. Advantages include:

- High FPS for real-time performance
- End-to-end differentiable network
- Global reasoning over entire image
- Multiple object class detection simultaneously

4.2 YOLO Detection Pipeline

- Input image is fed to the YOLO model
- Backbone CNN extracts features
- Image divided into grid; each cell predicts bounding boxes
- Low-confidence boxes removed
- Non-Maximum Suppression retains most confident predictions

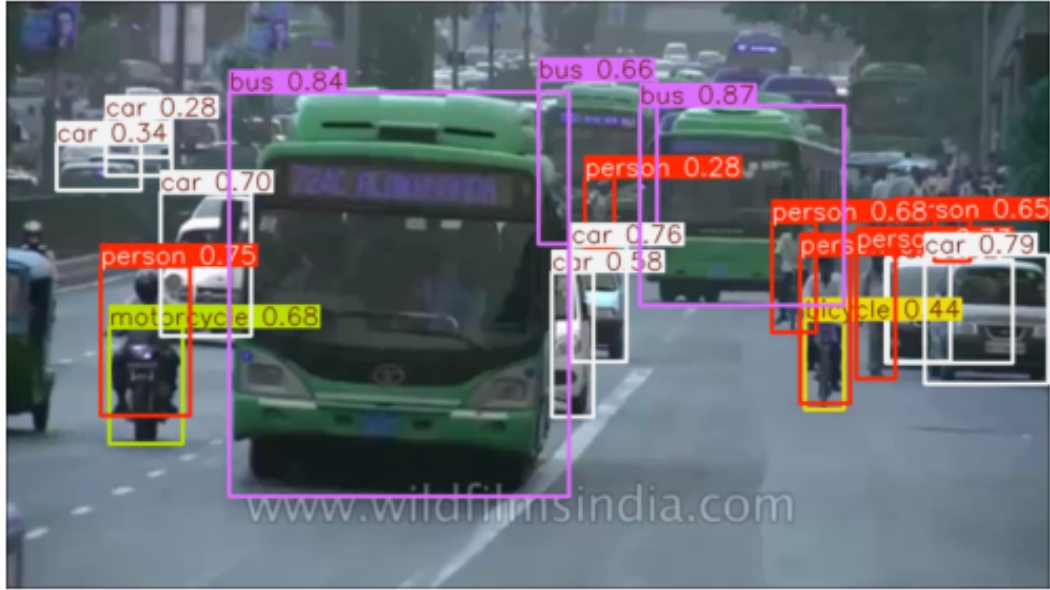


Figure 4.1: YOLO object detection output on road scene showing vehicles and pedestrians

4.3 Training Dataset

Annotated frames include:

- Vehicles: cars, buses, trucks
- Pedestrians
- Traffic signs

Training settings:

- Input image size: 416×416
- Batch size: 16
- Learning rate: 0.001
- Epochs: 50

4.4 Discussion

YOLO provides robust real-time object detection that, combined with lane detection, forms a reliable perception module for autonomous driving.

Chapter 5

Embedded Deployment

5.1 Model Optimization

The trained model is optimized for edge deployment using:

- Quantization
- TFLite conversion for Raspberry Pi deployment

5.2 Real-Time Inference

The optimized model runs on Raspberry Pi, enabling real-time lane and object detection with minimal latency.

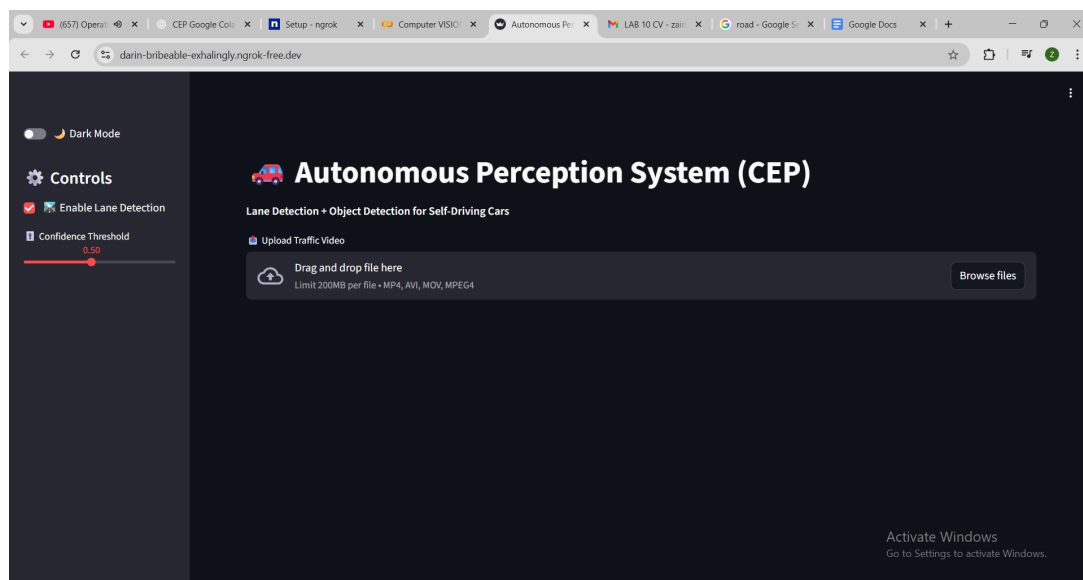


Figure 5.1: Streamlit UI visualization showing live lane and object detection

Chapter 6

System Evaluation and Discussion

6.1 Performance Metrics

The system was evaluated for:

- Accuracy
- Frames Per Second (FPS)
- Inference latency
- CPU, GPU, and memory usage

6.2 Results

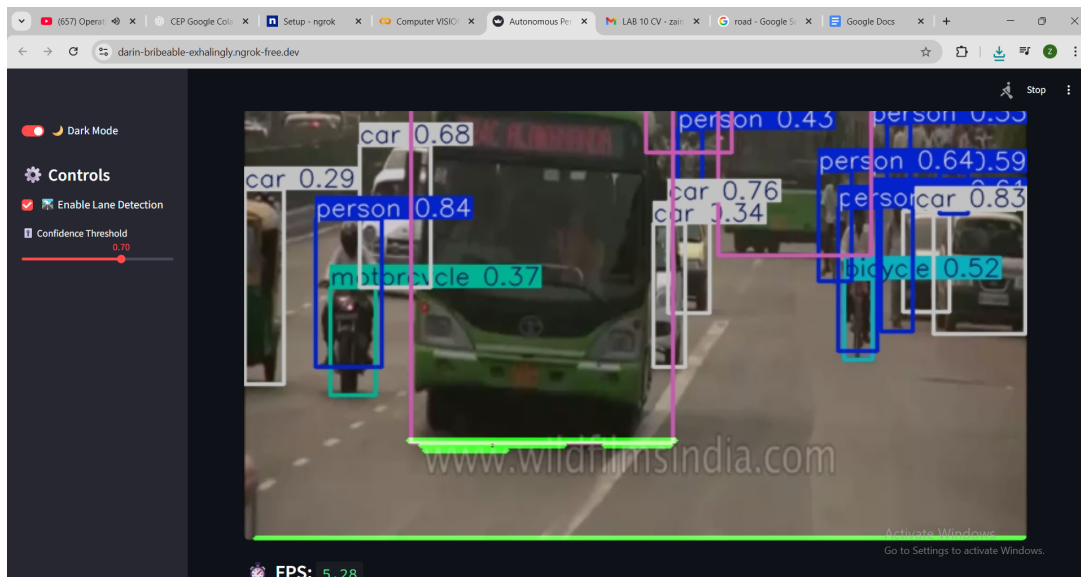


Figure 6.1: Performance metrics evaluation: FPS, accuracy, and resource usage

6.3 Discussion

The system successfully detects lanes and obstacles in real-time:

- Input images are correctly processed for lane and object detection
- Streamlit UI provides clear visualization
- Evaluation metrics confirm high FPS and accuracy

6.4 Limitations and Future Work

- Performance may degrade in extreme lighting or motion blur
- Future improvements: sensor fusion, advanced lane segmentation, optimized edge deployment

Chapter 7

Conclusion

This CEP demonstrates a complete workflow for vision-based autonomous perception using lane detection and YOLO object detection. The optimized model performs efficiently on edge devices and provides a reliable perception module for autonomous driving.

References

- Redmon, J., Farhadi, A., "YOLOv3: An Incremental Improvement", arXiv:1804.02767, 2018.
- Udacity, "Computer Vision for Self-Driving Cars", 2022.
- Raspberry Pi Documentation: <https://www.raspberrypi.org/documentation/>
- Lane Detection Tutorial, OpenCV Docs: <https://docs.opencv.org/>